Small Modular Reactor Facility-Class Design Basis

DBA-MA-SMR-FC1

Design Basis, Design Basis Accidents, and the Military-Action Extension — Near-Term Integral-LWR SMR Class

Issued under the DBA-MA Nuclear Series · IAEA Parent Framework (L1)

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Site-neutral facility-class Design Basis — written to stand on its own for any applicant, to be submitted for one-time regulatory review, and to be referenced by site-, national-, and contract-specific derivatives.

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ERRATUM TO VERSION 0.9

This version is the successor to Version 0.9 (version DOI 10.5281/zenodo.22730040; concept DOI 10.5281/zenodo.21430410), whose deposited text is not edited. Two passages of Version 0.9 §6.3 are in error against the October 2026 Edition of the Foundational Definitions, and the same correction reaches §§6.1, 6.2 and 6.4.

First, Version 0.9 §6.3 reads: “a change to a safety-significant nuclear process parameter that is initiated or materially influenced by a generative-AI system falls above the Bright Line and requires a qualified Command Broker to authorize it substantively before it takes effect.” That places an action above the line by the technology that originated it. Placement is on the consequence of the action (FD-BL §4.2), and material influence decides only whether the obligation reaches an action already placed above the line (FD-BL-D1 §7). Section 6.3 now reads that such a change falls above the Bright Line where the consequence of that change places it there.

Second, Version 0.9 §6.3 reads: “monitoring, advisory presentation, and deterministic machine-learning functions that are deterministic within a stated input bound and formally testable as deployed are outside Bright Line analysis under the carve-out and may operate without interposition.” Read as a list, it grants the carve-out by function, which FD-LD §4.3 forbids, and “may operate without interposition” restates a permission FD-LD §4.3 does not grant. Section 6.3 now states the carve-out as a property test, and places output of an in-scope monitoring or advisory component that makes no change of its own below the line as a process action carrying FD-BL-D2, following CB-Annex-Nuclear.

The same correction is carried into §6.1 and §6.4, where “generative-AI system” is replaced as the scope noun by “in-scope component” (FD-LD §4.3); into §6.2, where “may operate autonomously” is withdrawn; and into §6.4, where the human who defines the envelope is named as the Command Broker discharging in advance and accountability is allocated as FD-BL-D1 §8 allocates it. Nothing outside Section 6 changes in substance: the source term, the three-layer adversarial design basis, the passive-safety adequacy result and the 72-hour ASSC floor stand as deposited. A reader who reaches Version 0.9 through its version DOI will not see this erratum; it reaches the reader who follows the concept DOI.

1. Purpose and Scope

1.1 Purpose

This document establishes the Design Basis, the Design Basis Accident envelope, and the Design Basis Accident — Military-Action extension for the near-term integral light-water reactor small modular reactor (SMR) facility class. It is the first facility-class instrument in the DBA-MA nuclear series to treat the SMR as a distinct facility type requiring its own design-basis development. The large light-water reactor — the facility class implicit in the nuclear series since its inception — has a design basis established over decades of NRC regulatory practice. The SMR does not. This document begins to correct that absence, and it does so at class resolution: it is written to be site-neutral and technically complete, to stand on its own for any applicant, and to be submitted for one-time regulatory review as a topical report referenced by many subsequent applications.

The document serves three related purposes. First, it establishes the SMR-specific Design Basis — the envelope of conditions that SMR facilities must be demonstrated to withstand — and distinguishes that envelope from the large-LWR basis on which existing regulatory instruments were built. Second, it develops the non-adversarial Design Basis Accident envelope for the near-term integral-LWR SMR, accounting for design characteristics that differ materially from large-

LWR accident progression. Third, it extends both layers to the military-action threat tier and states, as a result rather than an open concern, where the SMR's passive-safety and source-term arguments remain adequate under deliberate action and where they do not.

The source term is the spine of this document. The licensing argument for reduced SMR emergency planning zones rests on the integral primary system's smaller source term and the passive safety system's capability to sustain safe shutdown without operator action or external power. That argument is analytically sound for the accident envelope. It is conditional — and therefore silent — on military action, because military action can do things accidents cannot: it can deny site access for the duration required to exhaust a passive system's design margins, it can target the shared infrastructure that serves multiple modules simultaneously, and it can sustain pressure on a recovery architecture that passive safety was not designed to defeat. Section 3A develops the source term; Section 5 applies the Conditionality Check that the cross-cutting instrument DBA-MA-ST requires of any consequence relaxation.

1.2 Scope: Near-Term Integral-LWR SMRs

This document addresses near-term integral light-water reactor SMRs as the initial SMR facility-class scope. Three design families are within scope: the NuScale VOYGR (approximately 77 MWe per module, up to twelve modules per campus), the Holtec SMR-300 (approximately 300 MWe per module), and the GE-Hitachi BWRX-300 (approximately 300 MWe per module, a boiling-water variant). These designs share three characteristics that define the scope boundary: integral or compact primary-system architecture that substantially reduces large-bore primary piping; passive safety systems capable of achieving and maintaining safe shutdown without active pumping, external power, or operator action for a defined duration; and active NRC design certification or pre-application review as of the date of this document. All three employ conventional low-enriched uranium fuel in the standard LWR range.

The following SMR families are explicitly outside the current scope: molten-salt reactors, high-temperature gas-cooled reactors, sodium-cooled fast reactors, and lead-cooled fast reactors. These have materially different thermal-hydraulics, fuel-cycle characteristics, passive-safety architectures, and source-term profiles — including, for several, the use of high-assay low-enriched uranium (HALEU). Their DBA-MA analysis requires facility-class-specific development that the near-term LWR framework cannot carry without distortion. Subsequent FC instruments will address each as their licensing pathways mature. This scoping decision is analytical, not evaluative.

Floating and underground SMR installations are noted as emerging configurations whose DBA-MA profiles differ materially from the land-based campus addressed here. They are identified in the facility-class register (Section 3.3) as deferred scope, with the rationale for deferral stated.

1.3 The Reference Deployment

A representative near-term integral-LWR multi-module campus — sited inland, within a weakly-interconnected or islanded transmission market, with access to an industrial water supply — is used throughout this document as the reference deployment. The reference deployment illustrates the methodology without limiting it: every general finding is demonstrated against it, and the methodology is nationally applicable without modification. Concrete national- and market-specific instantiations — for example, an ERCOT/Texas contract derivative — inherit from this class document by reference and carry the site-specific regulatory-coordination, licensing-precedent, airspace, and grid-interaction analysis that a named deployment requires. Those specifics belong to the derivative, not to the class document; this document states the class of gap and the general finding, and the derivative instantiates it.

The islanded-grid reference market is chosen because it provides the sharpest illustration of why SMR deployment is valuable from a grid-reliability standpoint and because its emergency-support topology is the most demanding: an islanded grid has no synchronous neighbor to supply automatic reserve at the speed and scale a synchronous interconnection provides. Section 2.4 develops the grid context; Section 5.5 develops the grid interaction under military-action conditions.

1.4 Relationship to Governing Documents

This document is issued under the DBA-MA Parent Framework (IAEA L1). The parent establishes the governing threat taxonomy, the four characteristics of military action, the consequence-analysis methodology, and the series architecture. This facility-class document inherits from the parent and applies those foundations to the SMR facility class without repeating them. It is also issued under the DBA-MA nuclear series: the IAEA L2 Accident Description, L3 Trigger Framework, and L4 Monitoring Guidance establish the phased accident-progression framework, the tiered response architecture, and the monitoring requirements that apply to SMR facilities as they do to large LWRs; where SMR-specific adaptations are required — passive-safety duration, multi-module campus considerations, reduced-staffing operational models — this document identifies them without displacing the governing L2–L4 instruments.

Four further instruments are load-bearing for this document and are referenced rather than restated. DBA-MA-ST (Source Term Under Military Action) states the Consequence-Basis Conditionality principle and the Conditionality Check applied in Sections 3A and 5. CB-Annex-Nuclear is the AI-governance binding case for the dual-DB stated in Section 6. DBA-MA-PPA supplies the physical-protection assessment methodology drawn on in Sections 4A and 5. DBA-MA-UxS-NA (the uncrewed-systems nuclear annex) supplies the standoff-barrier hardening measure cited in Section 7. Two further instruments substantiate the passive-safety adequacy result of Section 5A and are referenced there: DBA-MA-DiD (Defense-in-Depth Under Military Action) supplies the correlated-barrier adequacy method, and DBA-MA-SMR-FC1-TB1 supplies the thermal-hydraulic basis for the ASSC coping duration under simultaneous multi-module

actuation. Version numbers are not pinned in this document; each reference resolves against the owner's authoritative mirror in accordance with the portfolio version rule.

2. The SMR Facility Class

2.1 Defining Characteristics of the Near-Term Integral-LWR SMR

Five design characteristics define the near-term integral-LWR SMR as a facility class distinct from the large LWR and warrant its own design-basis development.

Integral or compact primary system. The near-term LWR SMR integrates the reactor core, steam generators, pressurizer, and primary coolant flow within a single pressure vessel or a compact arrangement that substantially eliminates large-bore primary piping. This eliminates the large-break loss-of-coolant accident — the governing design-basis accident for large LWRs — as a credible initiating event, and with it the fast early-release sequence. The non-adversarial DBA envelope for the SMR is therefore structurally different from the large-LWR basis.

Passive safety systems. Integral-LWR SMRs achieve and maintain safe shutdown through natural circulation, gravity-fed cooling, and passive heat removal that requires no active pumping, external power, or operator action to initiate. This supports the source-term relaxation argument for a reduced EPZ. It is also the characteristic the DBA-MA extension must examine most carefully: passive systems that win given time can be defeated by an adversary who systematically denies that time.

Factory fabrication and modular deployment. Near-term SMR designs are factory-fabricated and deployed as modular units, reducing construction time and quality-assurance variability relative to large-LWR field construction. For DBA-MA purposes, it also means module replacement — the ultimate recovery option after severe physical damage — has a defined supply-chain pathway that a sophisticated adversary can identify and potentially interdict.

Below-grade installation. Several near-term designs place the reactor module below grade, within a water-filled pool that provides passive decay-heat removal and physical protection against external events, including aircraft impact and standoff kinetic attack. Below-grade installation is a meaningful hardening characteristic. It does not eliminate the DBA-MA vulnerability at the campus level, because the shared infrastructure serving multiple below-grade modules — the switchyard, the cooling-water supply, the security and emergency-response organization — is not below grade.

Reduced operational staffing. SMR designs are licensed on the premise of reduced minimum staffing relative to large LWRs, reflecting reduced passive-safety surveillance complexity and a smaller consequence footprint per module. Reduced staffing is a grid-economics advantage. It is also the SMR's most acute DBA-MA vulnerability: when site access is denied, there is no staffing reserve to draw on.

2.2 The Multi-Module Campus — The Governing DBA-MA Configuration

The multi-module campus — four to twelve modules on a contiguous or proximate site, sharing common infrastructure — is the governing configuration for DBA-MA analysis. Single-module analysis understates the military-action risk for campus deployments in a specific and material way: campus-level shared infrastructure creates single points of failure that individual-module passive safety systems cannot overcome.

The shared-infrastructure elements that define the multi-module campus vulnerability are the utility switchyard and transmission interconnection serving the entire campus; the cooling-water intake and supply whose loss degrades passive heat removal across all modules simultaneously; the security-force organization whose staffing reduction under threat affects protection of all modules; the emergency-response organization whose minimum shift complement covers the entire campus; and the shared digital instrumentation-and-control (I&C) architecture whose common components create common-cause failure pathways across modules.

The concentration logic follows the same analytical pattern developed for hyperscale campuses in the DBA-ES-CL Critical Loads subsector (Project 8): individual modules are hardened, but the shared infrastructure they depend on is not equally hardened. An adversary who understands campus architecture will attack the shared infrastructure rather than the hardened individual modules. Installing modules below grade provides meaningful kinetic protection at the module level; it does not address the shared switchyard, the cooling-water intake, or the shared digital I&C architecture above grade.

The reference deployment is a multi-module campus of six modules with an aggregate net electrical output on the order of 300 to 450 MWe, depending on design family. This configuration is used throughout for consequence-chain analysis, regulatory mapping, and gap characterization.

2.3 SMR Deployment Typologies

Four deployment typologies are identified for the SMR facility class. The near-term scope of this document covers the first two; the latter two are registered as anticipated extensions.

Single-module pilot. One module serving as a first-of-kind deployment demonstrating the design in an operating environment. Single-module pilots have simpler shared-infrastructure profiles than multi-module campuses and are more closely aligned with the large-LWR single-unit case, within near-term scope.

Multi-module campus. Four to twelve modules on a shared campus, with common switchyard, cooling, security, and emergency-response infrastructure. The governing DBA-MA configuration. Within near-term scope; the reference deployment.

Remote and off-grid SMR. One to four modules operating islanded, without grid interconnection, serving a remote industrial or community load. The DBA-MA profile is

materially different: access denial is an inherent operating condition rather than an adversary-imposed one, and the recovery architecture must assume no external support. Deferred — requires separate FC development.

Industrial process-heat SMR. Co-located with an industrial facility to provide electricity and high-temperature process heat. The DBA-MA profile includes the industrial facility's own consequence chain as a coupled system. Deferred — requires separate FC development.

2.4 Grid Context: The Islanded-Grid Reference Market

An islanded or weakly-interconnected grid provides the richest context for demonstrating why near-term SMR deployment is valuable from a grid-reliability standpoint. Three characteristics motivate the deployment argument, and all three generalize beyond any single market.

Electrical isolation. An islanded grid has no synchronous interconnection to adjacent systems and only limited direct-current (HVDC) tie capacity. It is essential to state the physics precisely: an HVDC tie shares scheduled power between two asynchronous systems, not synchronous inertia — the two systems' frequencies remain independent, so a DC tie does not supply the frequency-stabilizing rotational energy that a synchronous interconnection provides. An islanded grid must therefore balance its own generation and load in real time with no automatic reserve-sharing backstop, which makes baseload generation that supplies genuine synchronous inertia uniquely valuable.

The inertia deficit. Large synchronous generators provide rotational inertia that resists frequency change in the seconds after a disturbance. As synchronous fossil units retire and are replaced by inverter-based resources, the inertia available per unit of installed capacity declines. Inverter-based resources — wind, solar, and battery systems — phase-lock to the grid but carry no rotating mass and provide no inherent inertial response; grid-forming inverter controls can synthesize a fast frequency response, but that is a controlled emulation, not stored rotational energy, and it depends on available headroom and functioning controls. A synchronous SMR supplies both the inertial response and the grid-forming voltage-and-frequency reference that an islanded grid locks onto. In the reference market, it is the anchor — the frequency buoy that ties the island together — and this role, not merely the megawatts, is what the grid most needs and cannot easily replace.

The baseload gap. As synchronous baseload retires, an islanded grid loses fuel-diverse, non-weather-dependent capacity at the same time it loses inertia. SMRs deployed into such a market are not simply adding generation; they are adding the kind of generation the grid most needs: synchronous, baseload, fuel-diverse, and available when wind and solar are not. A severe-weather event that produces simultaneous generating-unit trips — of which Winter Storm Uri (ERCOT, February 2021) is the validating reference — demonstrates the consequence of the inertia deficit: the loss of large synchronous generators, not merely a fuel-supply failure, is what

drove frequency toward collapse. An SMR fleet contributing synchronous baseload directly addresses that failure mode.

A concrete market instantiation — including named licensing precedents, a specific reliability coordinator, and the coupling to co-located large loads such as hyperscale data centers — belongs to the contract derivative, which reads the grid context from this section and adds the market-specific analysis. That coupling, addressed from the load side by the DBA-ES-CL series (Project 8), is not carried in this class document.

3. Design Basis: The SMR Facility Class

This section establishes the DB layer — the first stage in the DB → DBA → DBA-MA sequence. It defines why the SMR facility class is critical, what the design envelope is, what the facility-class register looks like within the SMR class, and what geopolitical siting assessment is required for deployments in geopolitically complex environments. Section 3A then develops the source term, which is the design-basis property that drives dose, the emergency planning zone, and the licensing pathway.

3.1 Why the SMR Facility Class Is Critical

The SMR facility class is critical to national and allied energy security for three independent reasons, each defining a distinct consequence pathway if SMR facilities are disrupted.

Distributed baseload for constrained grids. Large-LWR economics require interconnections capable of absorbing a significant loss of generation without instability. SMRs are sized for grid contexts where such interconnection does not exist — islanded topologies, remote industrial microgrids, and developing-nation grids whose load centers cannot support large baseload units. The SMR’s grid contribution is not simply megawatts; it is synchronous inertia, frequency stability, and baseload availability at a scale the grid can absorb and cannot easily replace.

Industrial decarbonization and process heat. High-temperature process heat from SMRs provides a pathway to decarbonizing industrial processes — hydrogen production, ammonia synthesis, chemical processing, desalination — that electrification alone cannot reach at current grid temperatures. The criticality argument here is economic and strategic: process-heat-dependent sectors constitute a decarbonization bottleneck whose resolution depends on SMR availability.

Allied energy security. SMR deployments in allied nations — Eastern European NATO members rebuilding energy infrastructure after dependence on adversary pipeline gas, Indo-Pacific partners seeking fuel-diverse baseload — reduce the energy-security leverage adversary energy exporters exercise. Disruption of SMR facilities serving allied energy security carries strategic consequences beyond the immediate grid-reliability impact.

3.2 The SMR Design Envelope

The SMR design envelope defines what the near-term integral-LWR SMR is built to do and the boundaries within which its safety systems are designed to function. It is the reference against which both the DBA event categories in Section 4 and the DBA-MA scenarios in Section 5 are assessed.

Thermal power range. Near-term integral-LWR SMRs operate from approximately 250 MWt per module (NuScale VOYGR) to approximately 900 MWt per module (BWRX-300). Campus aggregate thermal power in the six-module reference deployment is on the order of 1.5 to 5.4 GWt across design families, yielding roughly 300 to 450 MWe net — a significant generating block, but below the single-generator loss threshold that would instantaneously destabilize a well-formed grid.

Passive-safety / ASSC design-basis duration. Near-term SMR designs are licensed on the basis that passive safety can sustain core cooling and decay-heat removal without operator action, external power, or active pumping for a design-basis duration. The normative figure that governs DBA-MA analysis is the licensed minimum, not a vendor maximum claim. For the near-term integral-LWR class, the governing figure is 72 hours; Section 5A establishes this as the Autonomous Safe Shutdown Condition (ASSC) design-basis duration and treats design-family variation explicitly.

Reduced EPZ. NRC has approved a plume-exposure emergency planning zone at or near the site boundary (on the order of the site boundary to two miles for the leading near-term design) based on source-term analysis demonstrating that no credible accident sequence produces offsite doses exceeding the EPA Protective Action Guideline thresholds beyond that boundary under a range of meteorological conditions. This EPZ reduction is the regulatory consequence of the source-term relaxation developed in Section 3A; Section 5 examines its limits under military action.

Below-grade installation geometry. Several designs place reactor modules below grade in water-filled pools that provide passive heat removal, biological shielding, and physical protection against external events. The above-grade footprint is limited to reactor-building roof structures, auxiliary systems, and the interface with electrical and cooling infrastructure. This geometry concentrates the kinetic attack surface at the above-grade shared infrastructure rather than at the reactor modules themselves.

3.3 Facility-Class Register Within the SMR Class

The facility-class register identifies the major facility types within the SMR class that collectively constitute the scope of SMR-specific DBA development, and indicates which sub-classes have completed DB development, are in progress, or have not yet been addressed.

Sub-Class	Reference Design(s)	Status	Notes
Single-module pilot	VOYGR (1 mod), SMR-300, BWRX-300	Within scope — addressed here	Analytically simpler; large-LWR single-unit precedent partially applicable
Multi-module campus	VOYGR (4–12 mod)	Within scope — governing DBA-MA configuration; reference deployment	Shared infrastructure creates campus-level vulnerability absent from single-module analysis
Remote / off-grid SMR	Various — design-specific	Deferred — subsequent FC instrument	Access denial is an inherent operating condition; requires separate methodology.
Industrial process-heat SMR	Various — design-specific	Deferred — subsequent FC instrument	Coupled industrial-facility consequence chain requires separate treatment

3.4 Geopolitical Siting Assessment — Working Methodology

This section provides the working version of the geopolitical siting-assessment methodology registered as a gap in the DBA Series Gaps and Recommendations document. Because the SMR facility class is the series instrument most urgently dependent on that methodology — SMR licensing reviews citing source-term relaxation are active now — a working version is carried here rather than deferred to the future DBA-MA-SITE instrument. Findings derived from it are preliminary pending that instrument's completion. This siting overlay is distinct from the three-tier adversarial design basis of Section 4A: the siting overlay sets where on the threat-environment spectrum a site sits; the adversarial design basis sets the graded adversarial load the facility class must carry.

The assessment classifies a deployment site into one of three tiers based on the military-action threat environment. The classification governs the scope of DBA-MA analysis required before the source-term relaxation argument is applied to EPZ reduction.

Tier G1 — Stable, low kinetic threat. A stable allied nation with no credible near-term threat of military action against the site by a state or near-state actor. Tier 1 cyber threats are present in every connected environment regardless of geography; UAS threat from commercially available platforms is present; direct kinetic military action is not a credible near-term planning assumption. DBA-MA analysis at Tier G1 focuses on cyber-kinetic combined threats and UAS categories.

Tier G2 — Elevated geopolitical tension, limited kinetic threat. A nation experiencing elevated tensions with a state adversary, documented state-actor cyber pre-positioning, and a plausible but not imminent kinetic threat — NATO eastern-flank members in peacetime

conditions. Tier G2 requires full DBA-MA analysis across all four characteristics, with particular attention to precision and denial.

Tier G3 — Active conflict or high-probability near-term military action. A site within a nation experiencing active conflict or facing high-probability near-term military action. All four characteristics apply at maximum severity. The Zaporizhzhia nuclear power plant under military occupation is the reference case. Source-term relaxation arguments for EPZ reduction are analytically insufficient at Tier G3 without independent DBA-MA analysis demonstrating passive-safety adequacy for the full military-action consequence envelope.

A representative interior deployment in a stable allied nation is characterized as Tier G1 for kinetic threat and Tier G2 for cyber threat. This asymmetry — different classification for different threat categories at the same site — is the key methodological finding: geography reduces the kinetic threat, but it does not reduce the cyber threat, because cyber pre-positioning by Tier 1 state actors does not require geographic proximity to the target.

3A. The SMR Source Term — Technical Basis and Conditionality

The source term is the design-basis property that drives dose, the emergency planning zone, and the licensing-pathway gate. It is the spine of this document. This section develops the SMR-specific source term at class resolution and then declares the barriers on which the favorable result depends, in the form the cross-cutting instrument DBA-MA-ST requires. It does not re-derive the Consequence-Basis Conditionality principle — that is stated in DBA-MA-ST and referenced here; the former “source-term relaxation gap” carried in earlier revisions of this document is reclassified as the SMR-specific instance of that principle and cross-referenced accordingly.

3A.1 The Source Term as the Facility-Class Discriminant

The radiological source term is the quantity, composition, chemical and physical form, timing, and energy of the fission products available for release to the environment in a given sequence. It is not a single number but a release profile, and two features govern what follows. The first is the distinction between inventory and release fraction: the core inventory sets the total quantity of radionuclides present, while the release fraction — governed by the sequence and the barriers it encounters — sets how much of that inventory reaches the environment, in what form, and how quickly. The second is timing: a slow sequence allows short-lived, dose-dominant nuclides such as the radioiodines to decay, and allows particulates and vapors to deposit and plate out before release, whereas a fast sequence delivers a larger fraction of the inventory before either mechanism can act. A favorable source term is, in nearly every case, a statement that the sequence is slow and that the barriers governing release fraction hold.

3A.2 The Integral-Primary-System Source Term

The near-term integral-LWR SMR claims a mechanistic, design-specific source term in place of the prescriptive release fractions historically applied to large LWRs. The relaxation is earned, within the accident domain, by a genuine set of mechanisms. The integral primary system eliminates the large-break loss-of-coolant accident and with it the fast early-release sequence that drives large-LWR source terms. The large coolant-inventory-to-power ratio and wide thermal margin make sequences slow, buying the decay and deposition credit. And, in below-grade designs, the containment immersed in the reactor pool acts simultaneously as heat sink, retention barrier, and aqueous scrubber.

The class uses conventional low-enriched uranium in the standard LWR range. The favorable source term does not rest on any exotic fuel form; it rests on primary-system geometry and the immersed-containment barrier set. This is an important scoping point: the elevated-enrichment (HALEU) fuel forms that appear in the broader SMR field belong to the advanced non-LWR designs that Section 1.2 places out of scope, and the near-term integral-LWR source-term argument should not be conflated with them.

[Calibration item — resolved at v0.6] The design-specific radionuclide inventory, the credited decontamination factors for the immersed-containment water path, and the mechanistic release fractions are design-certification quantities that this class document characterizes but does not reproduce; each near-term design carries its own certified values. The class-resolution treatment states the mechanism and the conditionality; the normalized class-level framework consolidating the regulatory baseline and the per-design-family barrier set — against which each design's certified values attach — is provided in Section 3A.7.

3A.3 The Immersed-Containment Retention and Scrubbing Basis

In the below-grade designs the containment is immersed in the reactor pool, which extends the proven suppression-pool scrubbing mechanism by surrounding the whole containment in water. The pool provides three credited functions at once: it is the ultimate heat sink that keeps the sequence slow; it is a retention barrier outboard of containment; and it is an aqueous scrubber that strips iodine and particulate from any leakage path that passes through the water, at high decontamination factors. Each of these is a credited barrier, and therefore each is a conditionality: draining or bypassing the pool removes the scrubber and the heat sink together; a direct breach bypasses the retention credit; and persistence that sustains the initiating condition past the coping window erodes the slow-timing credit on which the decay and deposition depend.

3A.4 Dose, the Emergency Planning Zone, and the Licensing Gate

The favorable source term is what permits the contracted EPZ: the demonstration that no credible accident sequence produces offsite doses exceeding the EPA Protective Action Guideline thresholds beyond a boundary substantially smaller than the large-LWR ten-mile plume zone. The chain is source term → dose → EPZ → licensing pathway. Because the contracted planning zone is precisely what permits siting close to population and load, it also means the residual

adversarial scenarios that the accident source term does not bound have a nearer exposed population — so the favorable source term slightly raises the consequence of its own residual. This is not an argument against the design; it is a statement the Conditionality Check requires to be made rather than elided. The source-term risk profile also keys the licensing-pathway gate developed in Section 7: the gate is a function of consequence and fuel-mass risk profile, not of a megawatt band.

3A.5 Multi-Module Inventory Aggregation

The per-module source term is small, and this is a genuine and favorable property. It does not, however, extend without qualification to the multi-module campus. A campus aggregates radiological inventory across modules that share infrastructure, so single-module source-term logic does not carry to a correlated common-cause or adversarial multi-module event without separate justification. The governing source-term result at class resolution is therefore not the per-module figure but the aggregate available inventory of the campus under a correlated event that defeats shared barriers — the switchyard, the cooling-water supply, or the shared digital I&C — across multiple modules at once. This is the source-term quantity that Section 5 carries into the DBA-MA layer, and that Section 5A orders by source-term priority within the passive-safety adequacy result.

3A.6 The Conditionality of the Relaxation

The entire favorable source-term picture developed above is an accident-domain result. It is conditional on a specific set of credited barriers: pool water inventory, containment integrity, slow timing, and credited operator or passive mitigating action. Under the Consequence-Basis Conditionality principle stated in DBA-MA-ST, an accident-domain consequence relaxation does not bound the adversarial domain, because it is derived conditional on precisely the barriers that adversarial action is designed to defeat. By the scaling corollary, the SMR — which claims the largest relaxation of the nuclear facility classes — carries the largest adversarial residual relative to its claimed benefit, unless the credited barriers are independently protected against deliberate defeat.

This document accordingly adopts the Conditionality Check as a required element of its own analysis. The Check has three parts: declare the accident-domain assumptions on which the relaxation rests (done in Sections 3A.2–3A.5); test each credited barrier against the four characteristics of military action (Section 5.2); and state the residual — the consequence that remains credible when the credited barriers are deliberately defeated — and either show the barriers independently protected or carry the residual forward explicitly as a design-basis input (Section 5A). The former source-term relaxation gap is retired as a standalone open item and reclassified as the SMR-specific instance of the DBA-MA-ST principle.

3A.7 Class-Level Source-Term Reference

This subsection consolidates the class-level source-term reference flagged as a calibration item in Section 3A.2. It provides two references and, deliberately, not a third. The first is the regulatory baseline from which the near-term integral-LWR mechanistic source term departs: the prescriptive alternative-source-term release fractions of U.S. NRC Regulatory Guide 1.183, Revision 1, reproduced by radionuclide group in Table 3A-1. The second is the normalized per-design-family barrier set that governs how each certified design earns its mechanistic reduction from that baseline, given in Table 3A-2. The third — the design-specific radionuclide inventories, credited decontamination factors, and mechanistic release fractions — is not reproduced, because those are design-certification quantities carried by design-family reference; the class document fixes the structure and the public baseline, and each applicant's certified basis supplies the numbers.

Table 3A-1 states the large-LWR prescriptive release fractions the mechanistic source term replaces. They are given for the gap-release and early-in-vessel phases, separately for the boiling- and pressurized-water families, and they are the envelope against which each near-term design's earned reduction is measured. Iodine-131 and cesium-137 — the halogen and alkali-metal groups — remain the dose-dominant nuclides, and the guide's convention that ninety-five percent of released iodine is transported as cesium iodide is carried unchanged.

Radionuclide group	BWR — gap	BWR — early in-vessel	PWR — gap	PWR — early in-vessel
Noble gases	0.008	0.96	0.022	0.94
Halogens (iodine)	0.003	0.54	0.007	0.37
Alkali metals (Cs, Rb)	0.003	0.14	0.005	0.23
Tellurium group	0.003	0.39	0.007	0.30
Barium, strontium	0.00	0.005	1.4×10^{-3}	0.004
Noble metals	0.00	2.7×10^{-3}	0.00	0.006
Cerium group	0.00	1.6×10^{-7}	0.00	1.5×10^{-7}
Lanthanides	0.00	2.0×10^{-7}	0.00	1.5×10^{-7}
Molybdenum	0.00	0.03	0.00	0.10

Table 3A-1. Prescriptive alternative-source-term release fractions (large LWR), NRC RG 1.183 Rev. 1, Tables 1 (BWR) and 2 (PWR) — fractions by radionuclide group and release phase. These prescriptive values are the baseline the near-term integral-LWR mechanistic source term departs from; they are not the SMR source term.

Table 3A-2 normalizes the barrier set that earns the reduction, by design family. The three near-term designs do not earn the favorable source term the same way, and the class document must not collapse them to a single mechanism. NuScale's containment is immersed in a below-grade reactor pool that acts as heat sink, retention barrier, and aqueous scrubber at once — the configuration Section 3A.3 develops. Holtec's SMR-300 places a dry METCON containment inside a threat-resistant enclosure and relies on an evaporative annular water reservoir for an effectively unlimited coping period, with retention resting on containment integrity rather than an inboard water scrubber. The BWRX-300 uses a dry steel-plate-composite containment with no suppression pool, and its favorable source term rests first on preventing core uncovery — isolation condensers sustaining decay-heat removal for seven days without AC power, peak cladding temperature held below 1204 °C — rather than on scrubbing a release that is not postulated to occur.

Attribute	NuScale VOYGR	Holtec SMR-300	GE-Hitachi BWRX-300
Reactor type / power	Integral PWR; ~77 MWe per module (up to 12 per campus)	PWR; ~300 MWe per module	Natural-circulation BWR; ~316 MWe / 870 MWt
Primary system	Integral vessel (steam generators and pressurizer	Compact PWR; large coolant-to-power inventory	Integral BWR; isolation condensers; no large

	internal); no large-bore primary piping		recirculation loops
Containment	Steel containment vessel immersed in a below-grade reactor pool	METCON steel-concrete containment within a threat-resistant enclosure; annular cooling-water reservoir	Steel-plate composite (SPC) dry containment; no suppression pool
Credited retention / scrubbing	Reactor-pool immersion: aqueous scrubber plus retention outboard of containment (high DF for iodine and particulate)	Containment integrity (dry inside); the annular reservoir is heat sink, not an inboard scrubber	Dry-containment integrity; the primary credit is no core uncovery (peak clad < 1204 °C)
Heat-sink / coping basis	Passive pool heat sink; passive decay-heat removal for a defined coping window	Evaporative annular reservoir — replenishable, effectively unlimited coping	Isolation Condenser System (three trains); ~7 days on two trains with no AC power; pools replenishable
Source-term method	Mechanistic, design-specific (replaces prescriptive AST)	Mechanistic, design-specific	Mechanistic, design-specific
EPZ posture (public claim)	Site-boundary EPZ methodology (ACRS-validated, 2022)	Vendor states no offsite EPZ required	Reduced / site-boundary EPZ sought (per licensing)
Inventory · DFs · release fractions	Design-certification reference (NuScale FSAR)	Design-certification reference (Holtec licensing basis)	Design-certification reference (GEH licensing basis)

Table 3A-2. Near-term integral-LWR SMR source-term barrier set, by design family (class reference). Attributes synthesized from public design descriptions and licensing materials. Numeric inventories, decontamination factors, and mechanistic release fractions are design-certification quantities carried by design-family reference and are not reproduced here. Emergency-planning-zone entries are vendor design-basis claims subject to NRC review, not regulatory determinations.

Public sources: NRC Regulatory Guide 1.183 Rev. 1; NuScale VOYGR emergency-planning-zone methodology (NRC/ACRS, 2022); Holtec SMR-300 design description (HTB-085); GE-Hitachi BWRX-300 General Description (005N9751).

Two consequences follow for the class design basis. First, the conditionality of Section 3A.6 is barrier-set-specific: because the credited barrier differs by family — aqueous pool scrubbing, evaporative-reservoir coping, or dry-containment no-uncovery — the Conditionality Check of Section 3A.6 and the declared residual of Section 5A.4 must be applied against each family's actual credited barrier rather than a single generic pool credit. A pool-drain residual is a NuScale residual; it is not the BWRX-300 residual, whose governing question is instead the denial of isolation-condenser makeup past the seven-day coping window. Section 5A's source-term-prioritized response ordering therefore resolves differently by family, and a derivative that names a design inherits that family's barrier column together with its residual.

Second, this consolidation resolves the Section 3A.2 calibration item at class resolution and narrows the Section 9.2 open item. The class document now fixes the normalized framework — the regulatory baseline, the radionuclide-group structure, and the per-family barrier set against which certified values attach — and the design-family-referenced numeric inventories and release fractions attach to that framework through each applicant's certified basis rather than being reproduced here. What was a class-level source-term table to be consolidated is now the consolidated framework of this subsection plus a bounded numeric-attachment task carried at derivative resolution.

4. Design Basis Accidents: Non-Adversarial Envelope

This section establishes the DBA layer — the second stage in the DB → DBA → DBA-MA sequence. It defines the non-adversarial accident categories that bound the SMR design envelope, independent of adversarial cause. Each category is defined by its initiating event, its

consequence envelope for the near-term integral-LWR SMR, and its relationship to the large-LWR precedent. Section 4A then establishes the three-tier adversarial design basis, and Section 5 introduces military action as the initiating condition.

4.1 Passive-Safety System Design-Basis Events

4.1.1 Loss of Heat Sink — Single Module

Initiating event: loss of the ultimate heat sink for a single module — loss of feedwater flow, loss of condenser cooling, or loss of the passive heat-removal cooling pool — resulting in loss of the normal decay-heat removal pathway. This is the governing non-adversarial single-module event for the near-term integral-LWR SMR, replacing the large-break LOCA as the most consequential design-basis accident category for the integral primary system.

Consequence envelope: passive-safety actuation initiates natural circulation through the emergency core-cooling system; decay heat is removed passively to the reactor pool; no operator action, external power, or active pumping is required; and core and fuel remain within design limits for the ASSC design-basis duration. The envelope is substantially smaller than the large-LWR LOCA because the integral primary system eliminates the large-bore pipe break that drives large-LWR LOCA severity.

Relationship to large-LWR precedent: the large-LWR DBA envelope is driven by the large-break LOCA; loss-of-heat-sink scenarios drive the SMR DBA envelope. This is not a less severe accident category — it is a different governing event requiring different analytical methods and producing different consequence chains. This document identifies the category and its relationship to the DBA-MA extension without replicating the design-certification analysis.

4.1.2 Anticipated Transient Without Scram — SMR Characteristics

Initiating event: failure of the reactor protection system to insert negative reactivity following a transient initiator, requiring reliance on inherent reactivity feedback to arrest power increase. Near-term integral-LWR SMRs rely more heavily on inherent feedback — negative void, temperature, and in some designs power coefficients — than large LWRs of equivalent rod worth, and passive-safety actuation following initiation removes heat via natural circulation rather than high-pressure injection. The combined effect yields a more benign ATWS progression, consistent with the source-term relaxation. The DBA-MA consideration is whether cyber intrusion into digital I&C governing reactor protection could suppress the automatic protective functions that underpin this benign progression.

4.2 Multi-Module Common-Cause Events

4.2.1 Loss of Shared Switchyard

Initiating event: failure of the common switchyard serving all modules, causing simultaneous loss of offsite power to all modules. This is the governing common-cause non-adversarial event for the multi-module campus and has no equivalent in single-module or large-LWR single-unit analysis. The shared switchyard is the point of maximum concentration vulnerability: it is above grade, it is the campus's only connection to the bulk electric system, and its loss simultaneously initiates passive-safety actuation across all modules.

Consequence envelope: all modules simultaneously transition to passive-safety operation. The aggregate decay-heat load requires that the passive cooling pool maintain adequate heat removal for each module independently — satisfied by design for each module in isolation but requiring explicit verification for the simultaneous multi-module case (Section 8). Staffing for monitoring simultaneously actuated passive systems exceeds the minimum shift complement for a single-module event. For an islanded reference market, loss of the shared switchyard also disconnects the campus from the grid, dropping the campus's synchronous generation from the system in a single event.

4.2.2 Loss of Shared Cooling-Water Supply

Initiating event: loss of the common cooling-water intake or supply serving the campus — from drought, infrastructure failure, or disruption of the municipal or industrial water supply. Several near-term designs rely on the reactor pool as the ultimate heat sink; pool inventory is replenished from the site water supply under extended operation, so a sustained loss of makeup degrades the passive heat-removal capacity available beyond the initial ASSC period. Consequence envelope: passive safety functions normally during the initial period using pool inventory; beyond the ASSC design-basis duration, pool-inventory depletion requires restoration of makeup or an alternative heat-removal arrangement.

4.2.3 Common-Cause Digital I&C Failure

Initiating event: failure of shared digital I&C components serving multiple modules — common historian servers, shared safety-system data networks, or common human-machine-interface platforms — causing simultaneous loss of automated control functions or operator visibility across modules. Campus digital I&C architectures that share components create common-cause failure pathways that a single-module design-certification basis does not fully address. This category is the non-adversarial analog of the cyber-kinetic DBA-MA scenario in Section 5.4: including it in the non-adversarial DBA establishes that common-cause digital I&C failure is within the design basis regardless of cause, and that the protective architecture must address it whether the initiating condition is equipment failure or deliberate intrusion.

4.3 External Events Under Relaxed EPZ Assumptions

4.3.1 Extreme Heat and Grid Stress

Initiating event: an extreme-heat event producing simultaneous peak electrical demand, reduced thermal efficiency of air-cooled auxiliary systems, and elevated cooling-water temperatures that affect passive heat-removal margins. This is the warm-side complement to the cold-side severe-weather analog (of which Uri is the reference): instead of unit trips from frozen equipment, the scenario involves degraded safety margins from elevated ambient and cooling-water temperatures coinciding with peak grid stress — and the grid-side consequence of a campus trip under peak demand is the loss of exactly the baseload inertia a stressed grid most needs.

4.3.2 Coastal Siting Boundary

The reference deployment's inland siting preference is motivated by this category. A coastal SMR campus faces the same hurricane and storm-surge exposure as a coastal large LWR and requires the same design standards for wind loading and storm surge. An inland siting preference moves the reference deployment outside the primary strike zone while maintaining transmission access. Coastal deployments are within scope but require additional analysis for hurricane-related initiating events not developed in this class document; a specific coastal market instantiation is a matter for the derivative.

4.4 Spent Fuel and Waste Management

Several near-term integral-LWR SMR designs have no separate spent-fuel pool in the conventional sense: spent fuel is stored in the reactor pool or in an integrated storage system within the reactor building. The DBA-MA consequence chain for spent fuel under military action must therefore be developed for the specific SMR storage configuration rather than derived from the large-LWR spent-fuel-pool analysis; the large-LWR thermal-progression use case (DBA-MA-UC-1) is the model for the analysis structure but not directly applicable to SMR storage geometry. A dedicated SMR spent-fuel use case is identified in Section 7. The source-term conditionality of the spent-fuel pool is the sharpest illustration of the Consequence-Basis Conditionality principle and is developed in full in DBA-MA-UC-1; this document references that treatment rather than reproducing it.

4A. The Three-Tier Adversarial Design Basis

This section establishes the graded adversarial design load the near-term integral-LWR SMR facility class must carry, resolved by modality and threat level. It sits between the non-adversarial DBA envelope (Section 4) and the state-actor military-action extension (Section 5). It is terrorist- and sub-state-framed and lives within the world of the 10 CFR Part 73 design-basis threat; the state-actor acts-of-war escalation above its High tier is the DBA-MA layer of Section 5.

4A.1 Purpose and Relation to the DBT and the Siting Overlay

Three distinct constructs are in play in this document and must not be conflated. The geopolitical siting overlay (Section 3.4, tiers G1–G3) sets where on the threat-environment spectrum a site sits. The three-tier adversarial design basis (this section) sets the graded adversarial design load by modality — cyber, physical, and uncrewed-aircraft-system — at Low, Elevated, and High. The four characteristics of military action (Section 5.2) describe how a state adversary defeats credited barriers at the DBA-MA layer. The adversarial design basis here is retained in every derivative of this document; a derivative may record a determination that scopes out the state-actor military-action escalation of Section 5, but it does not scope out the three-tier adversarial design basis, which corresponds to the sub-state and terrorist adversaries already resident in the Part 73 DBT world.

4A.2 The Modalities

Cyber. Intrusion into the facility's digital I&C and supporting enterprise networks, ranging from opportunistic commodity malware to persistent pre-positioning in shared safety-system data paths.

Physical. Kinetic, standoff, and insider threats to plant structures, systems, and components — from trespass and vandalism through armed sub-state assault to precision standoff or insider-enabled sabotage of a specific credited barrier.

Uncrewed-aircraft-system (UAS). Incursion by uncrewed aircraft, from careless hobbyist overflight through deliberate surveillance and small-payload delivery to precision multi-platform payload delivery against above-grade structures, systems, and components. The term "uncrewed" is used throughout; standoff hardening and the counter-UAS authority gap are addressed in Section 7.

4A.3 The Three Tiers

For each modality, the design-basis adversarial load is graded across three tiers. The tiers are cumulative: a facility designed to the High tier in a modality is designed to the Elevated and Low tiers in that modality as well.

Tier	Cyber	Physical	Uncrewed-Aircraft-System
Low	Opportunistic commodity malware; no ICS-specific capability or intent.	Trespass and vandalism; no capability against credited barriers.	Careless or hobbyist incursion; no payload or targeting intent.
Elevated	Capable non-state or criminal actor with ICS familiarity; targeted but not persistent.	Armed sub-state assault of DBT class; capability against selected barriers.	Deliberate surveillance or small-payload delivery against selected targets.
High	Tier 1 state-actor pre-positioning in campus-level	Precision standoff or insider-enabled sabotage of	Precision multi-platform payload delivery to above-grade SSCs;

Tier	Cyber	Physical	Uncrewed-Aircraft-System
	shared digital I&C; persistent, coordinated command capability.	a specific credited barrier.	coordinated with other modalities.

4A.4 Assignment to the Facility Class

The near-term integral-LWR SMR facility class carries the full three-tier load. Consistent with the kinetic-low / cyber-high asymmetry established in Section 3.4 for interior deployments in stable allied nations, the High tier in the cyber and UAS modalities is the governing design driver for the reference deployment: cyber pre-positioning does not require geographic proximity, and the dense-airspace environment of many candidate siting zones makes the UAS High tier credible. The multi-module campus shared-infrastructure concentration (Section 2.2) is the amplifier across all three modalities — the same shared switchyard, cooling-water intake, and shared digital I&C that define the common-cause non-adversarial events are the elements an adversary at the High tier selects. The physical-protection assessment methodology in DBA-MA-PPA governs the barrier-by-barrier evaluation.

4A.5 The Escalation Boundary to the Military-Action Layer

Above the High tier of the terrorist-framed adversarial design basis sits the state-actor military-action escalation: intentionality, persistence, precision, and denial applied by a state adversary, of the Zaporizhzhia and correlated-grid-attack class. That escalation is the DBA-MA layer developed in Section 5. The boundary is recorded explicitly rather than by omission: a contract or national derivative that reduces scope away from the military-action layer carries a recorded applicability determination stating that what is excluded is the state-actor acts-of-war escalation only — not the adversarial design basis of this section, which is always retained. Methodology governs first and the contract second: a statement of work may make a reduction allowable, but it cannot force one, and the reduction, where taken, is precise and recorded.

5. Military-Action Extension: DBA-MA for the SMR Facility Class

This section establishes the DBA-MA layer — the third stage in the DB → DBA → DBA-MA sequence. It applies the four characteristics of military action to the SMR facility class, executes the Conditionality Check that Section 3A adopted, calibrates the threat honestly against the reference deployment’s tier classifications, and identifies the specific analytical residual the source-term relaxation leaves open when military action is the initiating condition.

5.1 The Source-Term Relaxation Under Military Action

The licensing argument for SMR EPZ reduction is analytically sound within its domain, and Section 3A states why. That demonstration, however, is conducted entirely within the non-

adversarial accident envelope: the design-certification basis does not include a military-action threat category in its accident-sequence analysis, and the EPZ reduction is derived from accident source terms rather than military-action source terms. Under the Consequence-Basis Conditionality principle, military action can produce source terms the accident analysis does not bound, for reasons that follow directly from the four characteristics. An adversary can target the passive-safety infrastructure specifically rather than generating the accidental conditions those systems are designed to mitigate; can sustain the initiating condition beyond the ASSC design-basis duration, defeating the passive system by denying it the time it requires; can coordinate simultaneous events at the shared-infrastructure level that defeat multiple modules' passive systems at once; and can deny site access during the period when passive-system margins are being consumed. The relaxation does not fail because the accidents it analyzes are wrongly analyzed; it is silent because it does not analyze the military-action initiating condition. This document supplies that analysis.

5.2 The Four Characteristics Applied to the SMR Facility Class

5.2.1 Intentionality

An adversary targeting an SMR campus does not attack randomly. It studies the campus architecture, identifies the shared-infrastructure elements whose loss has consequences across all modules simultaneously, and selects the configuration yielding maximum consequence with minimal operational complexity. The below-grade installation that the source-term analysis treats as a protective characteristic is, from the adversary's perspective, a reason to attack the shared infrastructure above grade rather than the protected module below it: the module's protective characteristic does not translate to a protective characteristic of the campus.

5.2.2 Persistence

The passive-safety system wins if given time — the ASSC design-basis duration is a genuine protective capability. An adversary who understands this will design the denial-of-access operation to exceed that duration; the attack need not damage the passive system directly, only deny the access that would restore active cooling, supply makeup, and let operators respond after the ASSC period has elapsed. The ASSC design-basis duration is therefore simultaneously the passive system's greatest strength and the adversary's most precise target: an adversary calibrating the denial period beyond it has, in effect, defeated the passive system without attacking it.

5.2.3 Precision

Modern state actors can execute precision attacks against specific above-grade infrastructure elements. At a Tier G1 site the precision kinetic threat is lower than at Tier G2 or G3, and it takes two credible forms: a precision UAS attack using commercially available or modified platforms capable of delivering a payload to a specific above-grade element, and a precision

cyber intrusion targeting specific digital I&C components identified through pre-positioning. The UAS precision threat is addressed further in Section 7; the cyber precision threat in Section 5.4.

5.2.4 Denial

The reduced-staffing operational model that makes SMR economics viable is the SMR's most acute denial vulnerability: a campus with a shift complement sized for passive-safety surveillance rather than active-safety operation has no equivalent to the operational reserve a large LWR's minimum staffing provides. At a Tier G1 site, denial takes the form not of military occupation but of area-wide emergency conditions — a wildfire, a chemical release from an adjacent facility, or a grid emergency — coinciding with a cyber intrusion that amplifies the denial of access. The adversary does not need military force to prevent access; it needs to create conditions in which the already-minimal shift complement cannot be augmented or relieved for longer than the ASSC design-basis duration.

5.3 The Reference Threat Calibration

An honest characterization of the reference deployment's DBA-MA threat environment distinguishes among three categories that do not share the same profile.

Kinetic threat — Tier G1. Direct military action against an interior campus in a stable allied nation by a state or near-state actor is not a credible near-term planning assumption. The DBA-MA analysis does not ignore kinetic threats — commercially available UAS, criminal or extremist actors with explosive devices, and the insider threat are all within the physical-protection design basis — but the Tier 1 precision kinetic-strike scenario that governs Tier G2 and G3 analysis is appropriately treated as a lower-probability planning input at Tier G1, stated explicitly rather than by applying an overstated threat.

Cyber threat — Tier G2. The Tier G2 cyber classification applies regardless of the site's kinetic tier. Tier 1 state actors conduct persistent cyber pre-positioning in critical-infrastructure networks without geographic discrimination; an interior campus's digital I&C is as accessible over the network as that of a frontier campus. The geopolitical threat reduction from interior siting applies to kinetic threats, not cyber threats — the key finding of the split.

UAS threat — dense-airspace complexity. The commercially available UAS threat is present at every SMR site. Where candidate siting zones carry dense legitimate UAS traffic — agricultural, inspection, surveillance, and delivery operations — a counter-UAS capability must discriminate threat platforms from legitimate traffic within an airspace-management framework not designed for nuclear-facility protection, and its coordination requirements are correspondingly harder. The specific dense-airspace instantiation for a named market is a matter for the derivative; the regulatory gap it exposes is addressed in Section 7.

5.4 The Multi-Module Campus DBA-MA Scenario

The governing DBA-MA bounding event for the reference deployment is a cyber-kinetic combined scenario at the campus level, calibrated to the Tier G1 kinetic and Tier G2 cyber environment. It does not require a state military presence; it requires a Tier 1 state actor's cyber pre-positioning capability, patience, and the ability to time execution to exploit grid-stress conditions.

The scenario proceeds as follows. A Tier 1 state actor has spent months pre-positioning within the campus's digital I&C architecture through the enterprise IT network, mapping the shared components serving all modules, identifying the historian servers and safety-system data networks whose manipulation would suppress alarms across the campus, and establishing a persistent presence capable of executing a coordinated command sequence on demand. The actor executes during a grid-stress period. Simultaneously, a commercially available UAS delivers a payload to the above-grade shared switchyard. The cyber component suppresses protective-relay actuation that would isolate the switchyard, extends the duration of the grid disturbance affecting the campus, and suppresses the SCADA alarms that would otherwise alert the shift complement before passive-safety actuation initiates across all modules.

Modules simultaneously actuate passive safety. The shift complement — sized for normal operations, not for simultaneous multi-module passive-safety monitoring — is immediately overextended, and the cyber component has corrupted the process-historian data that would normally allow remote monitoring from the operations center. The grid, already stressed, experiences the campus's synchronous generation dropping in a single event, degrading reserve margin and heightening the urgency of a grid emergency response concurrent with the nuclear emergency response. This is not the worst imaginable scenario; it is the worst credible scenario a reasonable analyst would identify within the range the reference deployment may encounter, and it is the bounding event that drives the design basis and the regulatory-gap arguments. Its named-market instantiation — a specific reliability coordinator, a specific severe-weather peak — is carried by the derivative.

5.5 Grid Interaction Under Military-Action Conditions

The bounding scenario results in the loss of the campus's synchronous generation into a stressed islanded grid. This is a grid-reliability event as well as a nuclear-safety event, and the two are causally linked in a way that the FERC Order 706-B NRC/NERC interface framework, designed for large-LWR single-unit events, only partially addresses. The grid-reliability consequences propagate through an islanded topology without the frequency-stabilizing backstop of a synchronous interconnection; under severe-weather grid stress, the simultaneous multi-module generation dropout may trigger automatic under-frequency load shedding, extending consequences beyond the campus fence line. The physics matters: the campus supplied genuine synchronous inertia and a grid-forming reference — the anchor role of Section 2.4 — and its loss removes both the stored rotational energy and the reference the island was locking onto, precisely when the stressed grid most needs them.

The institutional gap this creates is addressed in Section 7. Where the grid-reliability authority is a state body rather than FERC and NERC — as in an intrastate islanded market — Order 706-B supplies the conceptual model for the NRC / grid-reliability-authority interface but not the specific coordination framework, which does not currently exist and must be developed by the derivative for its named market. The coupling to co-located large loads (for example, hyperscale data centers whose workloads can migrate off-grid under stress, producing a compound bidirectional frequency excursion) is a real reference-market phenomenon; it is owned on the load side by the DBA-ES-CL series (Project 8) and carried by the contract derivative, not by this class document.

5A. The Passive-Safety Adequacy Result

This section states, as a result rather than an open concern, where the passive-safety architecture of the near-term integral-LWR SMR facility class is adequate and where it is not. It establishes the Autonomous Safe Shutdown Condition (ASSC) design-basis duration as a physical-DB result, discharges the Conditionality Check's residual requirement, and supplies the source-term-priority ordering that the dual-DB of Section 6 relies on. The result is expressed in the form the correlated-barrier method of DBA-MA-DiD defines — the credited barriers are tested for adversary-selected joint defeat rather than random coincident failure — and the coping-duration barrier, whose thermal-hydraulic basis under simultaneous multi-module actuation is established in DBA-MA-SMR-FC1-TB1, is the governing element under persistence and denial.

5A.1 The Autonomous Safe Shutdown Condition (ASSC)

The ASSC is the physical-DB condition in which the facility sustains safe shutdown and decay-heat removal without operator action, external power, or active pumping, for a design-basis duration. For the near-term integral-LWR SMR facility class this document carries a 72-hour ASSC design-basis duration as a conservative class floor, anchored to the established post-Fukushima station-blackout and FLEX coping benchmark that the passive designs are required to meet or exceed without operator action. The 72-hour figure is a floor, not a certified duration: the family-specific certified coping durations are the governing values for a specific application, and they are not uniform — large shared-pool designs credit boil-off coping measured in weeks, while isolation-condenser designs credit multi-day per-module durations. Two points govern its use in DBA-MA analysis. First, the coping duration credited for a multi-module campus is the aggregate simultaneous-actuation duration — the duration sustained when every module actuates at once against the shared ultimate heat sink — not the single-module certified duration, however large that single-module figure is. Second, the adequacy of the floor under simultaneous multi-module actuation is established in the technical-basis supplement DBA-MA-SMR-FC1-TB1, which reduces the question to whether each governing shared resource was sized for the aggregate N-module demand. For cross-class context, the large light-water reactor carries an ASSC design-basis duration on the order of two years; the SMR's 72-hour floor is materially

shorter and is the figure to which the AI-governance design basis of Section 6 keys its locked-envelope duration.

5A.2 Adequacy in the Accident Domain

Within the non-adversarial DBA envelope (Section 4), passive safety is adequate: for the ASSC design-basis duration, core and fuel remain within design limits without operator action, external power, or active pumping, and the source-term relaxation of Section 3A is earned. This is stated as a result, and it is the result on which the reduced EPZ and the licensing pathway rest.

5A.3 Adequacy in the Adversarial Domain, Bounded

Against the three-tier adversarial design basis (Section 4A), passive safety is adequate up to the ASSC design-basis-duration boundary. Within the 72-hour window, an adversarial action that does not defeat a credited barrier and does not sustain denial past the window is survived by the passive architecture without operator action. The below-grade geometry and the immersed-containment barrier set raise the effort required to defeat the credited barriers, which is a real and important protective margin against the Low and Elevated tiers in the physical and UAS modalities.

5A.4 Where Adequacy Fails — The Declared Residual

Passive safety is not adequate, and this document declares the residual rather than assuming it away, in three cases. First, persistence beyond the ASSC design-basis duration: an adversary who sustains denial of access past 72 hours has exceeded the passive-safety design basis, and the residual is the degraded-cooling source term that develops once the ASSC margin is expended under sustained denial. Second, correlated multi-module actuation: the aggregate ultimate-heat-sink load and inventory consumption of a simultaneous multi-module event differ from the single-module basis on which the certified coping durations are derived. The technical-basis supplement DBA-MA-SMR-FC1-TB1 closes this in method — reducing adequacy to verification that each governing shared resource was sized for the aggregate N-module demand — leaving a residual that is open in data only, until the certified shared-resource sizing basis is confirmed for the specific design. Third, direct barrier defeat: precision or standoff action that drains or bypasses the reactor pool, breaches containment, or severs the makeup pathway deletes the credited barrier and, with it, the source-term relaxation, and the residual is the deleted-relaxation source term aggregated across affected modules against a nearer exposed population. These residuals are carried forward as design-basis inputs; the protective requirement is to protect the credited barriers independently against deliberate defeat, or to accept the residual explicitly, but not to assume it away.

5A.5 Source-Term-Prioritized Response Ordering

Within the ASSC, control priority follows the source term: the deterministic control system protects in source-term priority order, not in order of operational or economic weight. This ordering is the physical-DB expression of the same rule the AI-governance design basis applies to criticality-band assignment (Section 6), and it is not an inheritance from the accident domain — under the Conditionality Check the priority is set by the inventory available once the credited barriers are defeated, which is what makes source term, rather than generic consequence magnitude, the governing discriminant. The 72-hour survivable-lock reconciliation that the AI-governance binding case defers to this facility-class document is discharged here: the SMR locked envelope must be survivable for the 72-hour ASSC duration, which imposes materially lighter endurance demands on the deterministic envelope than the two-year large-LWR case, and this difference is a design input to the envelope, the lock, and the hardware-reset gating described in Section 6.

5A.6 The Normalized ASSC Design-Basis Duration (Family-Accommodating)

Section 5A.1 carries the 72-hour ASSC design-basis duration as a conservative class floor. This subsection states the floor in normalized form, so that the family-specific certified coping durations are accommodated explicitly rather than folded into a single class number. For a campus of N near-term integral-LWR SMR modules, the ASSC design-basis duration is the aggregate simultaneous-actuation coping duration — the duration sustained when all N modules actuate passive safety at once against their shared resources — bounded below by the 72-hour FLEX-anchored class floor and governed, above that floor, by the family-specific certified duration evaluated on the aggregate N-module demand. The single-module certified duration, however large, is not the governing value; the technical-basis supplement DBA-MA-SMR-FC1-TB1 (§4.3) reduces the campus question to whether each governing shared resource was sized for the aggregate demand, and the declared residual of Section 5A.4 carries the case in which it was not.

The certified durations are not uniform, because the ultimate-heat-sink architecture that sets them differs by family. The supplement’s architecture taxonomy (TB1 §3) distinguishes three types across the near-term scope, and each locates the multi-module aggregation concern in a different shared resource. Table 5A-1 maps the taxonomy to the near-term design families and states the certified-duration character explicitly, with the numeric certified values carried by design-family reference as elsewhere in this document.

Architecture type (TB1 §3)	Reference design family	Governing thermal coupling	Certified coping duration — character (single module)	Where the campus aggregation concern lives
Type A — Shared immersed pool	NuScale VOYGR (all modules in one below-grade pool)	Aggregate pool boil-off — all modules reject decay heat to one inventory	Boil-off measured in weeks, then transition to air cooling — well above the floor per module	Sizing basis of the shared pool inventory: aggregate N-module versus per-module
Type B — Per-module isolation-condenser pools	GE-Hitachi BWRX-300	Per-module isolation condenser to a dedicated pool; shared makeup extends coping	Multi-day per module without makeup (≈7 days on two of three ICS trains, no AC power)	Sizing of the common makeup supply for simultaneous N-pool refill
Type C — Passive air-/evaporative-cooled containment	Holtec SMR-300 (evaporative annular reservoir)	Evaporative rejection to the atmosphere — a non-depletable ultimate	Effectively unlimited once replenished (water-augmentation phase, then	Augmentation-inventory sizing if shared; aggregate

		sink	evaporative / air rejection)	plume effect on the cooling path

Table 5A-1. Normalized ASSC coping-duration floor, by ultimate-heat-sink architecture and design family. Architecture types and governing couplings per TB1 §3–§4; family assignments and durations synthesized from public design descriptions (NuScale VOYGR; GE-Hitachi BWRX-300 General Description; Holtec SMR-300 HTB-085). Numeric certified coping durations and shared-resource sizing bases are design-certification quantities carried by design-family reference. The 72-hour class floor is the conservative lower bound anchored to the FLEX benchmark; the governing value for a campus is the aggregate simultaneous-actuation duration (TB1 §4.3).

Two uses follow. First, the residual of Section 5A.4 is read against this table: for Type A the residual is shared-pool boil-off consumed faster than the certified single-module basis under aggregate heat; for Type B it is the denial or under-sizing of the common makeup supply past the per-module pool duration; for Type C it is exhaustion of a shared augmentation inventory or degradation of the aggregate air-cooling path — each a different failure of the same floor, and each pointing at a different shared resource to protect. Second, the AI-governance design basis of Section 6 keys its locked-envelope endurance to the class floor, not to the family-certified duration: the deterministic envelope must be survivable for the 72-hour floor because that is the duration guaranteed across the class, while any margin a longer-coping family carries above the floor is a design input to that family's derivative, not a relaxation of the class-level lock.

6. The Dual-DB, Stated in Full

This is the document in which the dual-DB architecture is stated in full for the SMR facility class. The physical design basis is defined here; the AI-governance design basis is named and referenced to its binding case (CB-Annex-Nuclear), not redefined. The constructs owned by the AI Governance standards track — the Command Broker, the Bright Line, and the criticality-band model — are referenced by owner and are not restated.

6.1 The Two Design Bases

A nuclear facility deploying an in-scope component — a component that does not meet the determinism carve-out as FD-LD §4.3 states it, of which a generative-AI advisory or control-integrated system is the principal case — carries two design bases simultaneously. The physical design basis governs its reactor and safety systems and is defined in Sections 3 through 5A of this document. The AI-governance design basis governs the conditions under which such a component may influence the plant's process parameters. The two are independent in origin but interact at the DBA level, and the interaction sharpens at the DBA-MA level, where military action stresses both at once — the physical DB defining what must be protected and in what priority, the AI-governance DB defining who may act and under what enforcement.

6.2 The Physical-DB Half: The Determinism Posture

The discriminator for the safety instrumentation-and-control path is determinism and formal verifiability across the full credible input domain — the reactor-protection-system and safety-instrumented-system bar — not the "machine learning" label. This distinction is a physical-DB property and is in scope here. Frozen, fully verified machine-learning artifacts are admissible on

the safety path: once frozen and formally verified over the credible input domain, such an artifact is verified deterministic logic with machine-learning provenance, and it is governed by the same requirements as any other safety-path logic. Such an artifact meets the determinism carve-out and is outside the Command Broker obligation, which is a scope finding and not a grant of autonomy (FD-LD §4.3). Generative AI is excluded from the safety envelope (placement per FD-BL; the scope boundary is governed by FD-LD §4.3, by reference). The algorithm-drift rule follows directly: only frozen artifacts qualify for a safety role, and any retraining triggers full requalification — a retrained artifact is a new artifact and inherits none of the prior verification. This posture is grounded in the determinism and independence requirements of IEEE 603 and the digital-asset protections of 10 CFR 73.54. The rapid protective actions — reactor trip and engineered-safety-feature actuation — are already automated deterministic functions and are not generative-AI actions. They are unaffected by this posture and are outside the Command Broker obligation by the FD-LD determinism carve-out, because their behaviour is bounded by proof. They are not below the line: they are among the highest-consequence actions in the plant, and asserting they are below the line would be false. Judgment is not required where proof is available (FD-BL §4.5).

6.3 The AI-Governance-DB Half: Named and Referenced

The AI-governance design basis is governed by the Command Broker requirement, which is an obligation of the AI-governance design basis and not of the physical design basis. Its nuclear binding case is CB-Annex-Nuclear, the nuclear-sector annex to the cross-sector Command Broker framework; this document references that annex and does not redefine its constructs. In summary, and by reference: a change initiated or materially influenced by an in-scope component falls above the Bright Line where the consequence of that change places it there, and it then requires a qualified Command Broker to authorize it substantively before it takes effect; material influence decides whether the obligation reaches a change, and never whether the change is above the line (FD-BL-D1 §7). A component meets the carve-out by its properties — deterministic within a stated input bound and formally testable as deployed — and never by its function, so a monitoring or advisory function meets it only where it has those properties; a component that meets it is outside Bright Line analysis, which is a scope finding, not a permission, and not a finding that its actions are below the line (FD-LD §4.3; FD-BL §4.4). Output of an in-scope monitoring or advisory component that would make no change of its own is placed below the line as a process action and carries the indication-integrity obligation of FD-BL-D2 (FD-BL §4.2). Placement remains a question of consequence, determined action by action. The licensed Senior Reactor Operator is the natural locus of the Command Broker function for reactor process parameters, because that individual already holds the authority and accountability to authorize or refuse a change to a safety-significant parameter. Criticality-tier assignment follows the source term, consistent with the physical-DB ordering of Section 5A.5. The full placement of the Bright Line for nuclear process control, the qualification basis, and the accountability principle are established in CB-Annex-Nuclear and are not reproduced here.

6.4 The Interaction Point: ASSC Under Denial

The two design bases meet most sharply under military action, at the Autonomous Safe Shutdown Condition under denial. The governing determination is established in CB-Annex-Nuclear (the nuclear-sector instantiation of FD-BL-D1 — advance discharge only via a verified gatekeeper under R1–R3, with denied-condition immutability at §7) and referenced here: ASSC under denial is always discharged by deterministic and machine-learning systems operating inside a pre-authorized, human-defined envelope, and there is no denial condition under which an in-scope component receives autonomy above the Bright Line — denial does not relax the line, it removes the human who would otherwise stand on it, and the response is a deterministic envelope defined in advance by the Command Broker discharging in advance (FD-BR §4.2; FD-BL-D1 §7). On entry to the ASSC the machine-learning control is locked: the pre-authorized envelope becomes immutable for the duration of the denied condition (denied-condition immutability, FD-BL-D1 §7), executes in source-term priority order (Section 5A.5), and accepts no command path that would modify the envelope while the lock is in force — not from an in-scope component, not from an operator acting under coercion, and not from an adversary who has reached the control surface. Re-engagement is possible only through a hardware reset actuated by a restored Command Broker at the system itself, with no software, network, or remote release path. The SMR-specific reconciliation this document carries is the duration: the locked envelope must be survivable for the 72-hour ASSC design-basis duration established in Section 5A, which is the figure the binding case defers to the facility-class document. Accountability rests with the named Command Broker who authorized the envelope and accepted its proof, discharging in advance, and with the asset owner-operator, and returns to a present human on hardware reset; it never transfers to the system (accountability allocation per FD-BL-D1 §8).

6.5 Scope Boundary

This document defines the physical design basis in full and names the AI-governance design basis, deferring its full treatment to CB-Annex-Nuclear. The deepest AI-under-denial governance — the qualification methodology, the accountability principle, and the hardening design coordination for the envelope, the lock, and the hardware-reset gating — lives in CB-Annex-Nuclear and, for the broader artificial-intelligence quality-assurance and verification-and-validation treatment, in the TAF Nuclear Sector Application Guide. No Project 2 apparatus is restated here beyond the reference required to make the dual-DB complete.

7. Regulatory and Institutional Context

This section maps the regulatory instruments governing SMR deployment against the DBA-MA analytical requirements established in Sections 3 through 6. Each instrument is cited, credited for what it accomplishes, and assessed for the specific gap it leaves open for the SMR facility class.

It also states the source-term-keyed licensing-pathway gate and the presently-available UAS hardening measure.

7.1 The Licensing-Pathway Gate

The near-term integral-LWR SMR may be licensed under more than one pathway, and the governing discriminant is not a megawatt band but the source-term risk profile developed in Section 3A. 10 CFR Part 52 provides the established combined-license and design-certification pathway and is the route through which the leading near-term designs are pursuing NRC approval. 10 CFR Part 53, the risk-informed, technology-inclusive framework, offers a consequence-graded pathway. A proposed Part 57 — keyed to consequence and fuel-mass risk profile — would formalize a gate in which the licensing burden scales with the radiological inventory available for release and the barriers credited against it, rather than with rated electrical output. The point of stating the gate here is architectural: because the source term is the discriminant (Section 3A) and the source term's adversarial residual is what Section 5A declares, the licensing pathway should be selected by the same risk profile that governs the design basis, so that a design claiming a large consequence relaxation carries a correspondingly complete adversarial-domain demonstration before the relaxation is credited in its pathway.

7.2 NRC Part 52 — Combined License and Design Certification

What it accomplishes. 10 CFR Part 52 establishes the combined-license and design-certification pathways governing construction and operation of new nuclear facilities. Design certification provides site-independent NRC approval of a reactor design, referable in combined-license applications by multiple utilities, and requires a complete safety analysis report, probabilistic risk assessment, and physical-security plan for each application.

The gap. Part 52 design certification and combined-license review evaluate the design-basis accident envelope and the physical-security design-basis threat. They do not require a geopolitical situational assessment that evaluates the military-action threat category independently of the source-term analysis. The EPZ reduction approved under Part 52 is derived from accident source terms; the military-action source term — which Section 5 establishes is not bounded by the accident analysis — is not evaluated as part of the licensing basis. The geopolitical siting-assessment methodology of Section 3.4, and the Conditionality Check of Section 3A, fill this gap; incorporating them into the Part 52 process, as a required application element or as a generic communication establishing DBA-MA assessment as a supplemental requirement, is the regulatory action needed to close it.

7.3 FERC Order 706-B — NRC / Grid-Reliability-Authority Interface

What it accomplishes. FERC Order 706-B addresses the interface between NRC physical-security requirements and NERC Critical Infrastructure Protection reliability standards, establishing that nuclear plants are bulk-electric-system assets subject to NERC CIP

requirements coordinated with the NRC's 10 CFR 73.54 nuclear-cybersecurity requirements. It is the institutional model for the nuclear-safety / grid-reliability jurisdictional boundary.

Gap 1 — the state-jurisdiction interface. Wherever the grid-reliability authority is not FERC and NERC — as in an intrastate islanded market operating under a state public-utility commission — the FERC enforcement mechanism that gives Order 706-B its authority does not extend in the same way. Order 706-B supplies the conceptual model for the NRC / grid-reliability-authority interface but not the NRC / state-authority coordination framework such a deployment requires, and that framework does not currently exist. The class document states the gap as a class of gap; the derivative instantiates it for its named market and authority.

Gap 2 — the multi-module campus dropout scenario. Order 706-B was developed against the large-LWR single-unit model. A multi-module SMR campus presents a different profile: multiple modules actuating passive safety simultaneously produce a single-event generation dropout, in topologies where the dropout occurs without the frequency-stabilizing backstop of a synchronous interconnection. Whether the NERC reliability standards triggered by Order 706-B adequately address this scenario, or whether simultaneous multi-module actuation is a novel reliability-event profile the framework did not anticipate, requires explicit evaluation that has not been conducted.

7.4 10 CFR 73.54 and NEI 08-09 — Nuclear Cybersecurity

What they accomplish. 10 CFR 73.54 establishes the NRC nuclear-cybersecurity rule, requiring licensees to protect digital assets associated with safety, security, and emergency-preparedness functions through a defense-in-depth architecture and an NRC-approved cybersecurity plan. NEI 08-09 provides the industry implementation methodology — critical-digital-asset identification, attack-pathway analysis, security-control implementation, and compliance demonstration — and represents the professional consensus on adequate nuclear-cybersecurity implementation.

The gap for the SMR facility class. Both instruments were written against the large-LWR operating fleet. Three gaps follow. First, the critical-digital-asset identification methodology is structured around single-unit facility architecture, and campus-level shared components that constitute critical assets for multiple modules at once do not map cleanly onto it. Second, there is no campus-level implementation guidance for the shared digital I&C architecture serving multiple modules, with the common-cause cyber vulnerability that shared architecture creates. Third, the Tier 1 state-actor pre-positioning of the High cyber tier (Section 4A) is not addressed in a threat basis calibrated against a less-capable adversary model. An SMR-specific cybersecurity-plan guidance document, developed by NRC or NEI, is needed to close these gaps.

7.5 Counter-UAS Authority and Passive Standoff Hardening

What the current framework accomplishes. NRC physical-security requirements under 10 CFR 73 govern physical protection of nuclear facilities, including detection of unauthorized

aircraft and UAS; the UAS threat has been addressed through regulatory guidance and orders requiring licensees to detect and report incursions. Detection capability exists within the current framework.

The presently-available hardening measure. Detection without defeat is insufficient against the UAS threat at the DBA-MA bounding-event level, but a hardening measure is available now that requires no new authority. Passive kinetic standoff barriers for above-grade structures, systems, and components — the measure developed in DBA-MA-UxS-NA §8.2 — impose a standoff distance and a physical intercept on payload delivery to the above-grade shared infrastructure, and they can be deployed under existing physical-protection authority as a Tier-3 above-grade modification. They complement, rather than substitute for, active defeat authority: they raise the effort required for a precision UAS payload delivery against the switchyard or the cooling-water intake without depending on any statutory change.

The statutory gap. Active counter-UAS defeat authority is federal and does not currently extend to licensed commercial nuclear facilities. The FAA holds airspace jurisdiction under 49 U.S.C. § 40103, and DHS and DoD hold counter-UAS detect-and-defeat authority under Section 1602 of the FAA Reauthorization Act of 2018 and subsequent NDAA extensions, but that authority is limited to designated federal facilities. The needed provision is authority for nuclear-facility operators to deploy active counter-UAS defeat capability under defined threat conditions, modeled on the existing DHS authority and triggered by DBA-MA tier activation. The legislative vehicle has not been identified; this finding is carried into the legislative-outreach package developed from the DBA Series Gaps and Recommendations document. A dense-airspace market complicates the discrimination problem for any active system and is a matter the derivative develops for its named market.

8. Facility-Class Instruments and Inheritance

8.1 The Inheritance Model

This facility-class document is the parent from which site-, national-, and contract-specific derivatives inherit. A derivative declares what it inherits by reference to this class document and adds only what is locally specific — a named market, a named reliability coordinator, named licensing precedents, a named airspace environment — carrying a recorded applicability determination for any scope it reduces. Technical depth belongs here, in the class document; derivatives stay bounded and read from it. The contract derivative for a specific market (for example, the ERCOT/Texas SMR-licensing package) is the leading instance of this relationship: it inherits the design-basis characteristics, the source term, the three-tier adversarial design basis, the passive-safety adequacy result, and the dual-DB from this document, and instantiates the reference-market grid context, the state-authority coordination gap, and the dense-airspace UAS complexity for its named deployment.

Facility-class instruments issued under this document inherit its threat taxonomy, source-term treatment, adversarial design basis, consequence-analysis structure, and regulatory-gap characterization, and do not repeat them. Each adds what is specific to its analytical purpose.

8.2 Anticipated Instruments

DBA-MA-SMR-UC-1: Multi-Module Campus DBA-MA Consequence Trace. A use-case document developing the full consequence chain for the bounding DBA-MA scenario of Section 5.4 — the cyber-kinetic combined campus event under grid-stress conditions. The highest-priority instrument after this document.

DBA-MA-SMR-UC-2: SMR Spent-Fuel Thermal Progression. A use-case document addressing the spent-fuel-management consequence chain specific to near-term integral-LWR SMR storage geometry, for which the large-LWR spent-fuel-pool use case is not directly applicable.

DBA-MA-SMR-FC1-TB1: Multi-Module Simultaneous ASSC Adequacy (produced). The technical-basis supplement that substantiates the coping-duration barrier of Section 5A and closes the calibration item of Section 9.1, establishing the thermal-hydraulic adequacy of the ASSC floor under simultaneous multi-module actuation and reducing it to the shared-resource sizing basis.

DBA-MA-SMR-FC1-PILOT: Single-Module Pilot Supplement. An addendum addressing the simpler shared-infrastructure profile of a single-module first-of-kind deployment, in which the multi-module campus common-cause vulnerability is absent.

DBA-MA-SMR-POL: Policy Companion. A policy companion carrying the generic policy recommendations of this document to the audiences that must act on them. The grid-reliability-authority coordination framework identified in Section 7.3 is a market-specific matter carried by the contract derivative and coordinated with Project 5 (NERC regulatory implementation) and Project 8 (energy consequence envelope), not a class instrument.

8.3 Proposed Reference Events

This class proposes one reference event for *MA International Framework* (L1_DBA-MA_IAEA_Framework), acting as the nuclear sector framework, to confer under the author's determination of 14 October 2026 (*Energy and Nuclear Reference Events Follow the Same Method*, held in Project 0). It proposes the accident at the Fukushima Daiichi Nuclear Power Station, Japan, beginning 11 March 2011, for coping duration. The class's 72-hour floor for the Autonomous Safe Shutdown Condition (Section 5A.1) is anchored to the station-blackout and FLEX coping benchmark that the accident produced, and Section 5A.6 states that floor in normalized form. The accident is also the real event in which the conditions of the class's multi-module common-cause events occurred together. Several units on one site lost off-site power, their emergency power and their ultimate heat sink at once (Sections 4.2.1 and 4.2.2), a shared

structure carried the consequence of one unit into another, and access to the site was impeded during the period in which the units' margins were consumed, the same denial that Section 5.2.4 reads as an adversary's aim. The reading the instrument is asked to make is whether the class floor, and the aggregate simultaneous-actuation duration above it, would have held against the event's coping demand, and where the declared residual for persistence beyond the floor (Section 5A.4) would have been reached. The event is not adversarial, and the class proposes it as a non-adversarial reference for a duration that military action is designed to exceed.

MA International Framework confers the event at its version 2.2, in its Conferred Reference Events section at the close of its Section 11, as DBA-NUC-RE-2, on the author's decision of 15 October 2026. Its primary sector is Nuclear Reactors, Materials and Waste. The impacted sectors and the tier reading at conferral are the framework's and are recorded there. The tier determination is the instrument's, the instrument is Project 11's to write, and its recommendations return to this class as notes. Events at large light-water reactors, the seizure and occupation of the Zaporizhzhia plant among them (DBA-NUC-RE-1), are proposed by the framework in place of the owed large light-water reactor class, and are read against this class only where one of its provisions applies.

9. Gaps and Open Items

9.1 ASSC Design-Basis Duration — Multi-Family and Multi-Module Calibration

Section 5A carries 72 hours as a conservative class floor for the ASSC design-basis duration, anchored to the FLEX coping benchmark. The multi-module adequacy of that floor is addressed in the technical-basis supplement DBA-MA-SMR-FC1-TB1, which reduces the question to a finite set of shared-resource sizing-basis confirmations: the campus sustains the floor under simultaneous N-module actuation if, and only if, each governing shared resource — the shared ultimate-heat-sink inventory, the makeup supply, any coupled spent-fuel-pool load, and the air-cooling transition — was sized for the aggregate N-module demand rather than the single-module demand. The item is closed in method and open in data only: it resolves for a specific design once the certified inputs of the supplement's calibration register are supplied, and the residual case — a shared resource sized only for the single-module demand — is carried in the declared residual of Section 5A.4. The related class-level item — a normalized statement of the floor that accommodates the family-specific certified durations explicitly — is provided at v0.7 in Section 5A.6, which maps the TB1 ultimate-heat-sink architecture taxonomy (Type A shared immersed pool, Type B per-module isolation-condenser pools, Type C passive air-/evaporative-cooled containment) to the near-term design families and states the campus governing value as the aggregate simultaneous-actuation duration bounded below by the class floor. Section 9.1 is accordingly closed at class resolution; the remaining work is the design-specific numeric attachment carried by the supplement's calibration register.

9.2 Source-Term Technical Values — Design-Family Reference

The class-resolution source-term treatment of Section 3A states the mechanism and the conditionality; Section 3A.7 (added at v0.6) consolidates the normalized class-level reference — the RG 1.183 Rev. 1 large-LWR release-fraction baseline and the per-design-family barrier set against which each design earns its mechanistic reduction. The design-specific radionuclide inventories, immersed-containment decontamination factors, and mechanistic release fractions remain design-certification quantities carried by design-family reference and attach to that framework through each applicant's certified basis. The open item is accordingly narrowed from consolidating the framework — now done — to attaching the certified numeric values by design-family reference at derivative resolution.

9.3 Conditionality Check — Formal Registration

The Conditionality Check adopted in Section 3A is registered at series level as a formal cross-cutting requirement. DBA_Series_Gaps_Recommendations §1.8 (added at that document's v1.7) establishes the Consequence-Basis Conditionality Check as a mandatory cross-sector requirement for any facility class that claims a consequence relaxation, alongside the Military-Action Extension Applicability Determination of §1.7. With that registration, the reclassification of the former source-term relaxation gap as the SMR-specific instance of the DBA-MA-ST Consequence-Basis Conditionality principle is completed at the series level, and this item is closed. The SMR facility class (Section 3A.6, with the family-specific extension of Sections 3A.7 and 5A.6) stands as the first worked instance of the registered requirement.

9.4 Geopolitical Siting Methodology — Working-Version Status

The Tier G1/G2/G3 framework of Section 3.4 is a working version pending the DBA-MA-SITE instrument. The kinetic / cyber asymmetry is analytically sound; the specific boundary-classification criteria and the update process as geopolitical conditions evolve require the more developed methodology that instrument will provide.

9.5 SMR-Specific Nuclear Cybersecurity Plan Guidance — Not Yet Developed

The 73.54 and NEI 08-09 gaps identified in Section 7.4 — the absence of campus-level shared digital I&C guidance and High-tier state-actor calibration in the threat basis — require NRC or NEI action. This document characterizes the gap; closing it requires development of an SMR-specific cybersecurity-plan guidance document by the appropriate body. This finding should be transmitted to NRC as part of the regulatory-outreach sequence.

9.6 Counter-UAS Legislative Provision — Open Policy Item

The counter-UAS statutory gap of Section 7.5 is an open legislative item: authority for nuclear-facility operators to deploy active counter-UAS defeat capability under defined threat conditions, modeled on the existing DHS authority and triggered by DBA-MA tier activation. The passive standoff-barrier hardening of Section 7.5 is available now under existing authority and does not depend on this provision. The legislative vehicle is carried into the DBA Series Gaps and Recommendations legislative-outreach package.

9.7 Predictive UFLS Ride-Through Extension — Deferred to Position Paper

The grid interaction of Section 5.5 surfaces a grid-protection algorithm concept applicable across the bulk electric system: monitoring the observable pre-event indicators of an under-frequency event — declining reserve margin, generation-load imbalance, declining inertia — and temporarily extending the ride-through timer for high-value synchronous generators on local PMU parameters, so as to prevent the generator separation that converts a manageable frequency excursion into a cascade. The concept originates from this analysis but is deferred to a separate position paper, with NERC PRC-024 as the target for the standards revision; the SMR multi-module campus is the primary motivating use case. It is deferred pending completion of this facility-class document.

Revision History

Version	Date	Author	Description
0.1	May 2026	T. Roxey	Initial working draft. Establishes the three-layer DB → DBA → DBA-MA architecture for the SMR facility class. Scope limited to near-term integral-LWR SMRs. Regulatory citation architecture (10 CFR 73.54, NEI 08-09, NRC Part 52, FERC Order 706-B, counter-UAS authority) released for technical review.
0.2	May 2026	T. Roxey	Post-Grammarly revision. Added a second domestic nuclear licensing precedent and a hyperscale-load grid-coupling as reference-case material; added the predictive UFLS ride-through open item (deferred to a position paper).
0.3	July 2026	T. Roxey	Maturation into the site-neutral facility-class Design Basis. Executed the clean lift: the reference-market (ERCOT/Texas) material demoted from analytical spine to a single named illustration, with state-specific regulatory-coordination, siting-precedent, airspace, and workload-migration content relocated to the contract derivative. Added Section 3A (SMR source-term technical treatment) as the document spine, referencing DBA-MA-ST for the Consequence-Basis Conditionality principle and adopting its Conditionality Check; the former source-term relaxation gap reclassified as the SMR-specific instance of that principle. Added Section 4A (three-tier adversarial design basis: cyber / physical / uncrewed-aircraft-system at Low / Elevated / High). Added Section 5A (passive-safety adequacy result), establishing the 72-hour SMR Autonomous Safe Shutdown Condition (ASSC) design-basis duration as a physical-DB result. Added Section 6 (the dual-DB, stated in full): physical DB defined here, including the determinism / formal-verifiability AI/ML posture; AI-governance DB named and referenced to CB-Annex-Nuclear, not redefined. Added the source-term-keyed licensing-pathway gate (Part 52 / Part 53 / proposed Part 57) and UAS passive kinetic standoff barriers (from DBA-MA-UxS-NA §8.2) to Section 7 (regulatory context). Renamed retired DBA-DC references to DBA-ES-CL (Project 8) and relocated the workload-migration grid-coupling to the derivative. Reconciled stale version cross-references to name-only references that resolve against the owner's mirror. Grid-physics precision corrections applied throughout.
0.4	July 2026	T. Roxey	Incorporated two sibling instruments produced from this document's open items. Section 5A now cites DBA-MA-DiD (Defense-in-Depth Under Military Action) for the correlated-barrier adequacy method and DBA-MA-SMR-FC1-TB1 for the thermal-hydraulic basis of the coping-duration barrier. Section 5A.1 reframed: the 72-hour ASSC duration is stated as a conservative class floor anchored to the FLEX coping benchmark, with the family-specific certified durations named as the governing values

Version	Date	Author	Description
			and the aggregate simultaneous-actuation duration (not the single-module duration) identified as the figure credited for a campus. Section 5A.4 second residual updated to reflect TB1's method-level closure (open in data only). Section 9.1 rewritten: the multi-module calibration item is closed in method by reference to TB1 (adequacy reduces to shared-resource aggregate-sizing confirmation) and open in data only. Section 8.2 records TB1 as produced. Section 1.4 adds DBA-MA-DiD and TB1 as load-bearing references. No change to the source-term, adversarial-DB, or dual-DB sections.
0.5	August 2026	T. Roxey	FD-BL reconciliation: the two name-only references to the AI-Governance criticality construct updated to its reconciled name (criticality-band model / criticality-band assignment), matching TAF v2.6. No other content change.
0.6	August 2026	T. Roxey	Added Section 3A.7 (Class-Level Source-Term Reference), resolving the Section 3A.2 calibration item and narrowing the Section 9.2 open item. Consolidated the RG 1.183 Rev. 1 prescriptive release-fraction baseline (by radionuclide group, BWR and PWR, gap and early in-vessel) and a normalized per-design-family source-term barrier set — NuScale VOYGR pool-immersed aqueous scrubbing; Holtec SMR-300 METCON containment with an evaporative annular-reservoir coping basis; GE-Hitachi BWRX-300 dry steel-plate-composite containment with an isolation-condenser no-uncovery basis. Established that the Section 3A.6 Conditionality Check and the Section 5A.4 residual are barrier-set-specific by family; certified radionuclide inventories, decontamination factors, and mechanistic release fractions remain carried by design-family reference. Version references updated; no change to the passive-safety adequacy result or the dual-DB.
0.7	August 2026	T. Roxey	Added Section 5A.6 (The Normalized ASSC Design-Basis Duration, family-accommodating), completing the remaining class-level item of Section 9.1. States the ASSC design-basis duration in normalized form — the aggregate simultaneous-actuation coping duration, bounded below by the 72-hour FLEX-anchored class floor and governed above it by the family-specific certified duration on the aggregate N-module demand — and maps the TB1 ultimate-heat-sink architecture taxonomy (Type A shared immersed pool / Type B per-module isolation-condenser pools / Type C passive air-/evaporative-cooled containment) to the near-term families (NuScale VOYGR / GE-Hitachi BWRX-300 / Holtec SMR-300), with Table 5A-1 stating the certified-duration character and the campus aggregation concern per family. Section 9.1 narrowed to reference Section 5A.6 and closed at class resolution; the Section 5A.4 residual is now read by architecture type; the Section 6 lock endurance is confirmed keyed to the class floor. Certified coping durations and shared-resource sizing bases remain carried by design-family reference. No change to the source-term result or the dual-DB.
0.8	August 2026	T. Roxey	Closed §9.3: the Consequence-Basis Conditionality Check is registered at series level as a mandatory cross-cutting requirement in DBA_Series_Gaps_Recommendations §1.8 (that document advanced to v1.7), alongside the §1.7 MA Extension Applicability

Version	Date	Author	Description
			Determination, completing the series-level reclassification of the former source-term relaxation gap as the SMR-specific instance of the DBA-MA-ST principle. §9.3 body updated to reference the registration. No change to the technical content, the source-term result, or the dual-DB.
0.9	September 2026	T. Roxey	Placement corrected, not refined — the change reaches what this document permits, and a derivative author inheriting from it should read it as a correction. Section 6: the rapid protective actions (reactor trip, engineered-safety-feature actuation) are restated as outside the Command Broker obligation under the FD-LD determinism carve-out, because their behaviour is bounded by proof; the assertion that they sit below the Bright Line by their nature is withdrawn as false, those actions being among the highest-consequence in the plant (FD-BL §4.4, §4.5). The dual-DB summary is restated on the same basis: monitoring, advisory presentation and deterministic machine-learning functions deterministic within a stated input bound and formally testable as deployed are outside Bright Line analysis under the carve-out — a scope finding, not a permission, and not a finding that their actions are below the line — with placement left to consequence action by action. Formulation inherited from DBA-VC v0.9 rather than composed here. The document cited FD-BL §4.4 and FD-LD by reference at v0.8 while stating the placement incorrectly elsewhere; that condition is closed. Issued under Bright_Line_Placement_Conformance_Plan v0.4 (R-1, R-2). No change to the source term, the three-layer adversarial design basis, the passive-safety adequacy result, or the 72-hour ASSC floor.
0.9 (deposited)	September 2026	T. Roxey	Deposited to Zenodo under concept DOI 10.5281/zenodo.21430410 as version DOI 10.5281/zenodo.22730040, superseding the v0.5 deposit (10.5281/zenodo.21814883). No content change from v0.9 as issued; the citation block's version-DOI line is stamped with the reserved DOI and its claim that versions 0.7 and 0.8 were pending deposit is corrected — those versions were not deposited. This is the first deposit of this document since the 3 September placement correction, and it is that correction that reaches the published layer here.
0.10	October 2026	T. Roxey	Successor to the deposited v0.9 under CR-PB §6, carrying an erratum after the citation block that names the v0.9 version DOI (10.5281/zenodo.22730040) and the concept DOI and quotes the two passages in error; v0.9 is not edited. Conformed to the October 2026 Edition of the Foundational Definitions, on the Project 0 note of 2 October 2026 (finding S3-1) and of 3 October 2026 (Drift Screen finding P6-03), and following CB-Annex-Nuclear v0.4. §6.3: the reach sentence placed a change above the line by its origin in a generative-AI system and now places it on consequence, with material influence a test of reach (FD-BL-D1 §7); the carve-out clause, which read as a list granted the carve-out by function and restated "may operate without interposition", now states the property test and places output of an in-scope monitoring or advisory component that makes no change of its own below the line as a process action carrying FD-BL-D2 (FD-LD §4.3; FD-BL §4.2). §§6.1 and 6.4: scope noun "in-scope component" (FD-LD §4.3). §6.2: "may operate autonomously" withdrawn, the carve-out stated as a scope finding and not a grant of autonomy, and the scope boundary cited to FD-LD §4.3 in place of FD-BL §4.4. §6.4: the envelope author named

Version	Date	Author	Description
			as the Command Broker discharging in advance (FD-BR §4.2; FD-BL-D1 §7) and accountability allocated per FD-BL-D1 §8, the asset owner-operator included. Citation block restamped for v0.10, its version DOI to be assigned on deposit. Footer rebuilt: v0.9 carried the version and “Page 2” as one static run with no tab and no field, so every page read “Page 2”; it now carries a right-aligned tab and a PAGE field. No change to the source term, the three-layer adversarial design basis, the passive-safety adequacy result, or the 72-hour ASSC floor. Depositing this successor is the author’s act.
0.11	October 2026	T. Roxey	Section 8.3 (Proposed Reference Events) is added as the last subsection of Section 8, so that nothing is renumbered, under the author’s determination of 14 October 2026 (Energy and Nuclear Reference Events Follow the Same Method). The class proposes the accident at Fukushima Daiichi (11 March 2011) for coping duration, read against the 72-hour ASSC floor of Sections 5A.1 and 5A.6, the multi-module common-cause events of Sections 4.2.1 and 4.2.2 and the residual of Section 5A.4. MA International Framework confers it at its v2.2 as DBA-NUC-RE-2, on the author’s decision of 15 October 2026; the instrument is Project 11’s to write. Cover, citation block and footer read v0.11. The version DOI is still to be assigned on deposit, and v0.9 remains the deposited version this file supersedes; the erratum to v0.9 is unchanged. No change to the source term, the three-layer adversarial design basis, the passive-safety adequacy result, the 72-hour ASSC floor or Section 6. Depositing this successor is the author’s act.
0.10 (deposited)	October 2026	T. Roxey	Deposited to Zenodo on 3 October 2026 under concept DOI 10.5281/zenodo.21430410 as version DOI 10.5281/zenodo.23121885, superseding v0.9 (10.5281/zenodo.22730040). The file was uploaded without its citation block stamped, so the deposited PDF reads “not yet deposited”. This row records the deposit after the fact, on Project 0’s reading of the live record on 16 October 2026 (Note 2026-10-16c from Project 0).
0.11 (stamped for deposit)	October 2026	T. Roxey	Stamped for deposit on 16 October 2026 with version DOI 10.5281/zenodo.23420367, reserved by the author that day as a new version under concept DOI 10.5281/zenodo.21430410. No content change from v0.11 as issued. The citation block’s version-DOI line carries the reserved DOI and now names v0.10 (10.5281/zenodo.23121885), not v0.9, as the version superseded. The erratum to v0.9 is carried unchanged; v0.10 published it first. The 0.11 entry above, which says v0.9 is the deposited version this file supersedes, records what that revision held and is not edited.